

# IMPLEMENTATION SPRINT PLAN

## CBC Supervision Platform — Lot 1 delivery from V1.1 baseline

| Field                  | Value                                                                                  |
|------------------------|----------------------------------------------------------------------------------------|
| Document reference     | PLAN-CBC-IMPL-001                                                                      |
| Version                | 1.0 — Draft                                                                            |
| Date                   | 13 August 2026                                                                         |
| Parent                 | SPEC-CBC-UNIFIED-001 v1.0                                                              |
| Replaces for execution | Greenfield view of PLAN-SRV-MON-001 v2.1 (stories remapped)                            |
| Baseline               | Repository product V1.1 (enrolment, heartbeat, FastAPI, React dashboard, CBC Mail API) |

### 1. How to read this plan

- Sprints are labelled **FS0–FS8** (Future Sprints from current baseline).
- Story IDs: **FS{n} - {nn}** (e.g. **FS0-01**).
- Points: **1** = few hours · **2** ≈ 1 day · **3** ≈ 2–3 days · **5** ≈ 1 week.
- Roles: **S** senior · **J** junior/intern · **S+J** pair · **C** frontend.
- **DoD (all stories)**: code reviewed · unit tests where applicable · docs/DES updated if interface changes · demo'd in sprint review · requirement ID traced.
- **DoR**: SPEC ID + acceptance criteria + (for J) reference implementation pointer.

**Already Done (do not rebuild)**: enrolment token flow, basic heartbeat, FastAPI core API, PostgreSQL agents/alerts/users, React overview/agents/alerts/users/settings, JWT RBAC (3 roles), CBC Mail API + R11 channel health, availability windows / `machine_type`, WebSocket push, CSV export, packaging scripts (unproven silent CI).

### 2. Timeline overview

| Sprint | Duration  | Theme                               | Est. pts | Milestone                        |
|--------|-----------|-------------------------------------|----------|----------------------------------|
| FS0    | 2 weeks   | Contracts, design, inventory, stack | ~24      | M1 — Design & contracts approved |
| FS1    | 2–3 weeks | Plugin agent + resilience + DLQ     | ~28      | Plugin host live                 |
| FS2    | 2–3 weeks | Metrics depth + TSDB + packages     | ~31      | History & 3-OS install           |
| FS3    | 2–3 weeks | Log subsystem                       | ~27      | Logs searchable                  |
| FS4    | 2–3 weeks | Rules engine + webhooks             | ~32      | Alerting Lot-1 complete          |
| FS5    | 2 weeks   | Config groups + script coverage     | ~20      | Central config + AGT-013 mid     |
| FS6    | 2 weeks   | Reports, i18n, SNMP, connectors     | ~18      | Lot-1 perimeter closed           |

|     |           |                           |     |                            |
|-----|-----------|---------------------------|-----|----------------------------|
| FS7 | 2 weeks   | Stabilisation & NFR proof | ~15 | Ready for pilot            |
| FS8 | 2–3 weeks | Pilot, extinction, UAT    | ~12 | <b>M4 — Lot 1 accepted</b> |

**Indicative Lot 1 calendar:** ~18–22 weeks from FS0 start.  
**Lot 2 (FS9+):** separate plan after M4 (actions, n8n, OIDC, ITSM).

|     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |                 |   |      |                                         |
|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----------------|---|------|-----------------------------------------|
| FS0 | ■■■■ | FS1 | ■■■■ | FS2 | ■■■■ | FS3 | ■■■■ | FS4 | ■■■■ | FS5 | ■■■■ | FS6 | ■■■■ | FS7 | ■■■■ | FS8 (Lot 1 UAT) | ■ | ■■■■ | CBC Mail API kept; webhook added in FS4 |
|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----------------|---|------|-----------------------------------------|

### 3. Sprint FS0 — Contracts, design & foundations

**Goal:** Nobody blocked at FS1. Freeze schemas. Map PowerShell estate. Runnable stack with TSDB.

| ID     | Story                           | Acceptance criteria                                                                                                                                                                                 | Refs                     | Who | Pts |
|--------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----|-----|
| FS0-01 | Freeze canonical data contracts | Pydantic models <code>metric.v1</code> , <code>event.v1</code> , <code>task.v1</code> + plugin manifest in <code>shared/</code> ; JSON Schema exported; valid & invalid fixtures; published in repo | Part G, AGT-002, AGT-010 | S   | 5   |
| FS0-02 | Agent architecture dossier      | DES-001: component diagram, plugin lifecycle, collect & enrol sequences, agent state machine; editable + reviewed                                                                                   | DES-001                  | S+J | 3   |
| FS0-03 | Platform architecture dossier   | DES-002: components, data flow to UI, storage model, Compose/K8s deploy diagram                                                                                                                     | DES-002                  | S+J | 3   |
| FS0-04 | Interface dossier sync          | DES-003 reconciled with UI/UX Brief + current React screens; nav map; rights per role                                                                                                               | DES-003                  | J+C | 2   |
| FS0-05 | PowerShell coverage map v0      | DES-004 table: check → host(s) → planned plugin → status=planned; CBC ops workshop done                                                                                                             | DES-004, AGT-013         | S+J | 5   |
| FS0-06 | Compose stack with TSDB         | <code>docker compose up</code> starts Postgres + chosen TSDB + Redis + API; README one-command                                                                                                      | NFR-009, STO-001         | S   | 3   |

|        |                                  |                                                                                                                     |         |   |   |
|--------|----------------------------------|---------------------------------------------------------------------------------------------------------------------|---------|---|---|
| FS0-07 | Decide TSDB & log store          | Written ADR: VictoriaMetrics <b>or</b> TimescaleDB; Loki <b>or</b> OpenSearch; recorded in UNIFIED SPEC open points | Part L  | S | 2 |
| FS0-08 | CI lint/tests skeleton           | PR pipeline runs lint + unit tests (Linux required; Windows/macOS matrix planned)                                   | AGT-001 | S | 3 |
| FS0-09 | Agent simulator & contract tests | Simulator emits N hosts with valid/invalid payloads against frozen schemas                                          | —       | J | 3 |

**Demo:** DES artefacts presented; sample payload validated & stored; Compose up in < 5 min.

**Exit:** M1 approved; open points 1 & 1c addressed or scheduled with owners.

## 4. Sprint FS1 — Plugin agent core & resilience

**Goal:** Replace monolithic collector path with plugin host; durable buffer; schema-valid ingest + DLQ.

| ID     | Story                            | Acceptance criteria                                                                                                  | Refs                 | Who | Pts |
|--------|----------------------------------|----------------------------------------------------------------------------------------------------------------------|----------------------|-----|-----|
| FS1-01 | Plugin framework                 | Registry loads plugins by manifest; fake plugin runs; docs for authors                                               | AGT-002, AGT-080     | S   | 5   |
| FS1-02 | Single-instance lock             | Second process exits with clear error on Win/Linux/macOS                                                             | AGT-000, AGT-001c    | J   | 2   |
| FS1-03 | TLS verify + enrolment hardening | Agent rejects invalid certs (no <code>verify=False</code> in default prod mode); enrolment token single-use retained | AGT-003/004, SEC-001 | S   | 5   |
| FS1-04 | Durable store-and-forward        | Disk buffer default 24 h / 500 MB; ordered replay; zero loss after platform outage test                              | AGT-005, NFR-006     | S   | 5   |
| FS1-05 | Full identity on every payload   | AGT-015 fields + timezone on enrol and each metric/event batch                                                       | AGT-015              | J   | 2   |
| FS1-06 | Receiver schema validation + DLQ | Authenticated ingest; Pydantic <code>metric.v1</code> ; ACK; dedup by message ID; invalid → DLQ + alert              | PLT-001, PLT-002     | S   | 5   |

|        |                               |                                                                                                         |                  |     |   |
|--------|-------------------------------|---------------------------------------------------------------------------------------------------------|------------------|-----|---|
| FS1-07 | CPU golden-path plugin        | Manifest, collect (total+per-core), tests, docs — template for others                                   | AGT-020          | S+J | 3 |
| FS1-08 | Reserve task channel (reject) | Agent accepts <code>task.v1</code> envelope structure but L0 rejects execution with structured response | AGT-010          | S   | 2 |
| FS1-09 | Heartbeat/status regression   | Existing offline detection still works after refactor                                                   | AGT-006, ALR-002 | J   | 2 |

**Demo:** Kill API 10 min → buffer fills → restart → metrics catch up; second agent instance refused; invalid payload in DLQ.  
**Exit:** Plugin host is the only collection path for CPU.

## 5. Sprint FS2 — Metrics depth, TSDB, packaging

**Goal:** Full L0 system metrics in TSDB; silent packages; first PowerShell replacements.

| ID     | Story                     | Acceptance criteria                                                                                          | Refs                      | Who | Pts |
|--------|---------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------|-----|-----|
| FS2-01 | Memory plugin             | Matches golden path; OS-consistent values; unit tests                                                        | AGT-021                   | J   | 3   |
| FS2-02 | Disk plugin               | Per mount point; threshold-ready series                                                                      | AGT-022                   | J   | 3   |
| FS2-03 | Network + process plugins | Per-IF throughput/errors; Top-N + watched processes                                                          | AGT-023, AGT-024          | J   | 5   |
| FS2-04 | TSDB write + rollups      | Points written; rollups 1m→1h→1d; retention config 30d/13mo                                                  | STO-001, STO-002, PLT-011 | S   | 5   |
| FS2-05 | Host detail from TSDB     | Agent detail charts over selectable range                                                                    | DSH-002, DSH-006          | C+J | 5   |
| FS2-06 | Silent packages 3 OS      | CI builds <code>.deb</code> <code>.rpm</code> <code>.msi</code> <code>.pkg</code> ; silent install validated | AGT-012                   | S   | 5   |
| FS2-07 | First coverage plugins    | ≥ N PowerShell checks (from DES-004) delivered as plugins; map updated                                       | AGT-013                   | S+J | 3   |
| FS2-08 | Services/files collectors | Real collection for configured lists (not stubs); CBC list placeholders documented                           | AGT-026, CBC-AGT-03       | J   | 3   |

**Demo:** Charts for 24 h history; silent install on one Linux + one Windows host.  
**Exit:** STO-001 operational; coverage map has first delivered rows.

## 6. Sprint FS3 — Log subsystem

**Goal:** First-class log collection without a side shipper.

| ID     | Story                        | Acceptance criteria                                                  | Refs             | Who | Pts |
|--------|------------------------------|----------------------------------------------------------------------|------------------|-----|-----|
| FS3-01 | File log tailing             | Multi-file/glob; rotation; persisted offsets; no loss/dup on restart | AGT-030          | S   | 5   |
| FS3-02 | journald + Windows Event Log | System/Application (+ configurable channels)                         | AGT-031          | J   | 3   |
| FS3-03 | Parsers                      | regex/JSON/syslog; ts/severity/message; raw if parsed=false          | AGT-032          | J   | 5   |
| FS3-04 | Source filters               | include/exclude; drop stats reported                                 | AGT-033          | J   | 3   |
| FS3-05 | Multiline grouping           | Configurable start-pattern                                           | AGT-034          | S   | 3   |
| FS3-06 | Rate limit + spill           | Default 5 MB/min; local spill; platform alert on limit               | AGT-038          | S   | 3   |
| FS3-07 | Log store + UI search        | Full-text search; filter host/severity/period                        | STO-003, DSH-005 | S+C | 5   |
| FS3-08 | Pattern → event.v1           | Matching lines emit immediate events for ALR-003                     | AGT-036          | S   | 2   |

**Demo:** Tail nginx error log → appear in UI search < 60 s; rate-limit alert triggered in test.  
**Exit:** Log path Lot-1 Must accepted in review.

## 7. Sprint FS4 — Rules, alerts, notifications

**Goal:** Close Cahier Lot-1 alerting & outbound notify gaps.

| ID     | Story                          | Acceptance criteria                                                   | Refs             | Who | Pts |
|--------|--------------------------------|-----------------------------------------------------------------------|------------------|-----|-----|
| FS4-01 | Duration thresholds            | Rule “metric > X for N minutes”; editable in UI; no alert on spike    | ALR-001          | S   | 5   |
| FS4-02 | Four severities + auto-resolve | Info/Minor/Major/Critical; Open→Ack→Resolved; auto-resolve + timeline | ALR-004, ALR-005 | S+J | 5   |

|        |                            |                                                                      |                        |     |   |
|--------|----------------------------|----------------------------------------------------------------------|------------------------|-----|---|
| FS4-03 | Maintenance windows        | Per host/group; suppress + audit; distinct from availability windows | ALR-007                | J   | 3 |
| FS4-04 | Routing & escalation       | Recipients, channels, schedules, escalate after N min unacked        | ALR-006                | S   | 5 |
| FS4-05 | Keep CBC Mail API polished | Delivery status on alert; retry/backoff; templates                   | INT-001 (CBC), INT-009 | S   | 2 |
| FS4-06 | Signed HMAC webhook        | Per-rule URL; HMAC signature; retry; status visible                  | INT-003, INT-009       | J   | 3 |
| FS4-07 | Log-pattern alert rules    | ALR-003 wired to FS3 events                                          | ALR-003                | S   | 3 |
| FS4-08 | Alert UI polish            | Filters, ack, resolve, timeline, delivery status                     | DSH-004                | C+J | 3 |
| FS4-09 | Offline regression         | Existing ALR-002 + availability windows still correct                | ALR-002, CBC-AGT-02    | J   | 2 |

**Demo:** Sustained CPU 5 min → Major email + webhook; maintenance suppresses; spike does not alert.

**Exit:** Lot-1 notify channels = Mail API + webhook both green.

## 8. Sprint FS5 — Central config & coverage completion

**Goal:** Config no longer edited on each machine; PowerShell map largely delivered.

| ID     | Story                       | Acceptance criteria                                         | Refs    | Who | Pts |
|--------|-----------------------------|-------------------------------------------------------------|---------|-----|-----|
| FS5-01 | Config by machine group     | Create groups; assign agents; versioned config; rollback    | AGT-008 | S+J | 5   |
| FS5-02 | Agent applies remote config | Pull/push apply without SSH; ack version to platform        | AGT-008 | S   | 5   |
| FS5-03 | Coverage plugins complete   | All DES-004 planned → delivered (or waived with sign-off)   | AGT-013 | S+J | 5   |
| FS5-04 | CBC service/file lists live | Official lists from CBC loaded; collectors verified         | CBC-A/B | S   | 2   |
| FS5-05 | Agent footprint self-report | CPU/RAM of agent reported as metrics; alert if over AGT-007 | AGT-007 | J   | 2   |

|        |                          |                                                          |         |   |   |
|--------|--------------------------|----------------------------------------------------------|---------|---|---|
| FS5-06 | Duplicate-check flagging | Platform flags overlapping legacy script + plugin checks | AGT-014 | S | 3 |
|--------|--------------------------|----------------------------------------------------------|---------|---|---|

**Demo:** Change threshold for group “Agence” → agents pick up new version without local edit.

**Exit:** Central config is default ops path.

## 9. Sprint FS6 — Analysis, i18n, perimeter

**Goal:** Remaining Lot-1 Must visualisation & network perimeter.

| ID     | Story                     | Acceptance criteria                                          | Refs    | Who | Pts |
|--------|---------------------------|--------------------------------------------------------------|---------|-----|-----|
| FS6-01 | Custom dashboards         | User widget grids; shareable                                 | DSH-003 | C   | 5   |
| FS6-02 | Scheduled reports         | PDF/CSV schedule + on-demand                                 | DSH-007 | S+C | 5   |
| FS6-03 | i18n FR + EN              | Language switch; all Lot-1 screens                           | NFR-008 | C   | 3   |
| FS6-04 | SNMP/ICMP checks          | Poll network gear; status on dashboard                       | AGT-029 | S   | 5   |
| FS6-05 | Basic external connectors | At least one of: Docker host metrics or VMware/Hyper-V basic | PLT-004 | S   | 3   |

**Demo:** Custom board shared; EN UI; switch SNMP device appears offline.

**Exit:** Lot-1 functional perimeter declared complete pending FS7 proof.

## 10. Sprint FS7 — Stabilisation & performance

**Goal:** Prove NFRs before pilot.

| ID     | Story                 | Acceptance criteria                                      | Refs             | Who | Pts |
|--------|-----------------------|----------------------------------------------------------|------------------|-----|-----|
| FS7-01 | Load test 128→500     | Sustained load; no data loss; document results           | NFR-004, PLT-003 | S   | 5   |
| FS7-02 | Latency budgets       | Measure collect→UI ≤60 s; detect→notify ≤30 s; page ≤3 s | NFR-001/002/005  | S+J | 3   |
| FS7-03 | 3-OS regression suite | Automated smoke on Win/Linux/macOS agent builds          | AGT-001/012      | S   | 3   |
| FS7-04 | Backup/restore drill  | RPO ≤24 h demonstrated; runbook published                | STO-006          | S   | 2   |

|        |                          |                                        |         |   |   |
|--------|--------------------------|----------------------------------------|---------|---|---|
| FS7-05 | Hardening guides         | Agent + platform hardening docs        | SEC-008 | S | 2 |
| FS7-06 | Platform self-monitoring | Critical platform components monitored | NFR-010 | J | 2 |

**Demo:** Load report + restore drill in review.

**Exit:** Go/No-Go for FS8 pilot.

## 11. Sprint FS8 — Pilot, extinction, Lot 1 UAT

**Goal:** M4 — Lot 1 accepted; PowerShell scripts decommissioned where covered.

| ID     | Story                  | Acceptance criteria                                                                              | Refs                 | Who   | Pts |
|--------|------------------------|--------------------------------------------------------------------------------------------------|----------------------|-------|-----|
| FS8-01 | Pilot fleet onboarding | UAT family 1 on agreed pilot hosts                                                               | Part K               | S+J   | 3   |
| FS8-02 | Alerting E2E UAT       | Family 3 on Mail API + webhook                                                                   | Part K               | S+J   | 2   |
| FS8-03 | Resilience UAT         | Family 4 pass                                                                                    | Part K               | S     | 2   |
| FS8-04 | History/reporting UAT  | Family 5 pass                                                                                    | Part K               | J     | 2   |
| FS8-05 | Script extinction      | Each DES-004 row → verified in production then script decommissioned; zero uncovered Must checks | AGT-013/014          | S+CBC | 5   |
| FS8-06 | Lot 1 acceptance pack  | Traceability matrix requirement → test → evidence; sign-off                                      | UNIFIED SPEC 8.1.5e; | S     | 3   |

**Demo:** Coverage map at zero open Musts; UAT sign-off meeting.

**Exit:** M4 Lot 1 accepted.

## 12. Lot 2 backlog placeholder (FS9+)

Do **not** start before M4 unless explicitly approved.

| Theme                         | Refs                     | Notes                          |
|-------------------------------|--------------------------|--------------------------------|
| Action plugins + signed tasks | AGT-060–068, SEC-005/007 | Human approval, dry-run, audit |
| OIDC SSO + Security role      | DSH-025, API-003         | Corporate IdP open point       |
| n8n closed-loop pack          | N8N-001–007              | 5 starter workflows            |



|                                 |             |                  |
|---------------------------------|-------------|------------------|
| Noise suppression / correlation | ALR-008/009 | Tool choice open |
| ITSM / Teams / Slack            | INT-004/005 |                  |
| Scale / HA to 5 000             | NFR-003/004 |                  |

## 13. Capacity & sequencing rules

1. **FS0 is mandatory** before FS1 coding of plugins.
2. **FS3 (logs)** can partially overlap late FS2 if staffing allows; do not start before TSDB decision (FS0-07).
3. **FS4** depends on FS1 schemas + FS2 metrics (and FS3 for ALR-003).
4. **Senior-only zones:** FS1-03, FS1-04, FS3-01, FS3-06, FS4-01, FS7-01.
5. Update DES-004 at **every** sprint review until FS8-05.
6. CBC blockers (service/file lists, offline threshold, script inventory) escalate within 5 business days if unanswered.

## 14. Definition of Done — Sprint

- All committed stories Done or explicitly carried with reason
- No open Sev-1 defects in sprint scope
- UNIFIED SPEC status columns updated for touched IDs
- Demo recorded / notes filed
- Coverage map (DES-004) committed

## 15. First sprint commitment (FS0) — proposed board

### Committed (priority order):

1. FS0-01 Freeze contracts
2. FS0-07 ADR TSDB/log store
3. FS0-05 PowerShell map workshop + v0 table
4. FS0-06 Compose + TSDB
5. FS0-02 / FS0-03 DES dossiers
6. FS0-09 Simulator
7. FS0-04 / FS0-08 as capacity allows

**Owners to name:** Product (CBC), Tech lead (S), Intern (J), Frontend (C).

## 16. Change log

| Version | Date       | Changes                                                                   |
|---------|------------|---------------------------------------------------------------------------|
| 1.0     | 2026-08-13 | Initial implementation sprint plan from UNIFIED SPEC + remapped Plan v2.1 |

